

Ethics for Scientists and Engineers

Javier Larrosa

UPC

larrosa@cs.upc.edu

March 17, 2026

- **Sensitivity:** to raise awareness of ethical aspects of professional work
- **Knowledge:** to learn about professional standards
- **Judgment:** to develop skills in moral reasoning
- **Commitment:** to strengthen personal dedication to exemplary conduct

What does morality cover?

Think of something that you have witnessed (or suffered) that you consider unethical or morally wrong

- Why it was so?

Is Morality always about not harming?

Harmless Stories (J. Haidt, *The Righteous Mind*)

Is Morality always about not harming?

Harmless Stories (J. Haidt, *The Righteous Mind*)

- Eating your pet

Is Morality always about not harming?

Harmless Stories (J. Haidt, *The Righteous Mind*)

- Eating your pet
- Incest

Is Morality always about not harming?

Harmless Stories (J. Haidt, *The Righteous Mind*)

- Eating your pet
- Incest
- cleaning bathroom with flag
- ...

Is Morality always about not harming?

Harmless Stories (J. Haidt, *The Righteous Mind*)

- Eating your pet
- Incest
- cleaning bathroom with flag
- ...

Where moral values come from?

- These harmless morally wrong actions are probably evolution-driven
- Therefore, one could hypothesize that morality is a **human construct**
- and that it is a **pragmatical construct**
- This is an open controversial issue

In this course we will restrict to morality about actions where people can be hurt (in a broad sense) because (I think) we all agree in that causing harm is wrong

Outline of the following lectures:

- **Introduction** to Morality and Ethics
- Ethics for **Scientists**
- Ethics for **Engineers** (with emphasis on Informatics)

Outline of this lecture:

- Preliminaries
- Some Useful Tools and Definitions
- Ethical Theories

Morality vs Ethics

Morality

Collective Agreement about what is right and wrong

- Morality is about **shared principles**

Morality vs Ethics

Morality

Collective Agreement about what is right and wrong

- Morality is about **shared principles**

Ethics

Field of Philosophy about addressing morality in a critical objective way

- Ethics was triggered by the question that Socrates formulated 2500 years ago: how should we live? what is a good meaningful life?

Morality vs Ethics

Morality

Collective Agreement about what is right and wrong

- Morality is about **shared principles**

Ethics

Field of Philosophy about addressing morality in a critical objective way

- Ethics was triggered by the question that Socrates formulated 2500 years ago: how should we live? what is a good meaningful life?

Simplification

In everyday's life we often use **Ethics** and **Morality** as synonyms and in this course we will sometimes also do (i.e, morally wrong equivalent to unethical)

Morality and Religion

Along history, **Religion** has strongly influenced morality and **priests** have helped people to deal with it.

However, the **argumentation has often been weak**: *We have to behave in a certain way because this is what Gods wants us to do.*

Morality and Religion

Along history, **Religion** has strongly influenced morality and **priests** have helped people to deal with it.

However, the **argumentation has often been weak**: *We have to behave in a certain way because this is what Gods wants us to do.*

The contribution of Religion in morality and Ethics is very valuable. However, in this course we make a secular approach because we want to think critically and independently about morality

Morality without Religion

Moral principles are **behaviour guidelines** or **principles** that we all **voluntarily** follow because it **makes the world a better place for all of us**

Morality without Religion

Moral principles are **behaviour guidelines** or **principles** that we all **voluntarily** follow because it **makes the world a better place for all of us**

They really do?

- **Coordinated Intelligence** (in some cases being sub-optimal on our incentives is globally optimal, Tragedy of the Commons)
- **Civilization/humanity** (e.g. Human Rights)

Are moral principles immutable?

- In some cases morality follows **universal** principles
 - e.g.: killing

Are moral principles immutable?

- In some cases morality follows **universal** principles
 - e.g.: killing
- ...but in many other cases it is strongly based on context (culture, history)
 - e.g.: sexism (harassment, adultery), slavery,,...

Are moral principles immutable?

- In some cases morality follows **universal** principles
 - e.g.: killing
- ...but in many other cases it is strongly based on context (culture, history)
 - e.g.: sexism (harassment, adultery), slavery,...
- ... as a matter of fact the *no killing* universal moral rule is also adapted to context (e.g. killing a slave, self-defense, euthanasia, death penalty, abortion,...)

Are moral principles immutable?

Quizz

Say something that is acceptable now and you believe that it will not in, say, 100 years

Ethics helps us address morality in a rational way.

- Moral Judgments in complex situations
- Decide the right course of action

Ethics helps us address morality in a rational way.

- Moral Judgments in complex situations
- Decide the right course of action

A first ethical consideration: it is not the same,

- **Must do (moral imperative):** Help a person in need near you
- **May do:** donate to a charity

This is important, because people violating those rules deserve different moral judgments

Peter Singer Interview (mandatory to listen to)

Peter Singer is one of the world most influential ethicists.

In this interview Peter Singer discusses societal taboos and whether anyone is doing enough to make the world a better place.

Hammurabi Code (Babylon 1750 BC, now in Louvre)

- If a man accuses another man and charges him with homicide, but cannot bring proof against him, his accuser shall be killed.
- If a builder constructs a house for a man but does not make it conform to specifications so that a wall then buckles, that builder shall make that wall sound using his own silver.
- ...



Both want to encode rules for a civilized society

Both want to encode rules for a civilized society

Law:

- Must be as precise as possible
 - Difficult to adapt to changing environments (e.g. technology: online advertisement)
- Must be followed (it is enforced)

Both want to encode rules for a civilized society

Law:

- Must be as precise as possible
 - Difficult to adapt to changing environments (e.g. technology: online advertisement)
- Must be followed (it is enforced)

Ethics:

- Precision is not that important (e.g. lying is morally wrong?) because there is personal interpretation.
- Social pressure and self-esteem is what may enforce morality

Ethics vs Law

Both want to encode rules for a civilized society

Law:

- Must be as precise as possible
 - Difficult to adapt to changing environments (e.g. technology: online advertisement)
- Must be followed (it is enforced)

Ethics:

- Precision is not that important (e.g. lying is morally wrong?) because there is personal interpretation.
- Social pressure and self-esteem is what may enforce morality

First morality then Law

Laws are usually inspired by morality (Natural Law)

Why should we care about Morality?

It is not only about social recognition and/or having self esteem

A selfish reason

Morality as a social agreement means trust and trust makes society much more **efficient**

Morality can be seen as a *lubricant* for the *friction* of social interaction. Imagine,

- If we did not believe that other people respect traffic lights...
- If we could trust verbal agreements

Why should we care about Ethics?

If you do not care about Ethics, it means that you embrace the moral beliefs and norms given to you by your family and your society. You accept them without question or serious examination. In other words: you do not think critically about your moral principles.

Why should we study Ethics?



Groucho Marx (comedian)

Those are my (moral) principles, and if you don't like them... well, I have others

- Many studies on Psychology and Behavioural Economics show that we **align our values to our incentives** (e.g. experiment on exam cheating)
- So it seems a good idea to **anchor** our moral values as much as possible before we have a moral conflict

Why should we care about Ethics?

- ① Sometimes finding out the right thing to do is difficult
- ② Having some ethical background may help us deal with future ethical conflicts,
 - Help us identify/frame the conflict
 - Anchors our ethical standards
 - Gives us some rational background to deal with it

Disclaimer

Do not expect that studying Ethics will give you the answer to your ethical conflicts.

Ethics is a branch of Philosophy after all. Philosophy is more about framing questions than about giving answers.

- We have general moral principles, and sometimes we see that somebody, in a particular context, breaks them.
- Then we make a moral judgement...
- there are two very tempting **extreme approaches**:
 - Relativism
 - Absolutism

Relativism

All moral points of view are equally valid.

- What is right and wrong for one person is not necessarily so for another
- What is right and wrong depends on **context (time, place, culture,...)**
- Relativism is typical in open and **tolerant** environments

Relativism

All moral points of view are equally valid.

- What is right and wrong for one person is not necessarily so for another
- What is right and wrong depends on **context (time, place, culture,...)**
- Relativism is typical in open and **tolerant** environments

Problems:

- It invalidates moral judging
- It makes moral discussion impossible and meaningless

Relativism

All moral points of view are equally valid.

- What is right and wrong for one person is not necessarily so for another
- What is right and wrong depends on **context (time, place, culture,...)**
- Relativism is typical in open and **tolerant** environments

Problems:

- It invalidates moral judging
- It makes moral discussion impossible and meaningless

Example: Tolerating girls genital mutilation on the grounds of **cultural** respect

Example: Exonerating sexual harassment that took place **50 years ago**

Absolutism

There is a system of norms and values that should not be questioned

- **No exception:** a rule is a rule
- Typical in **authoritarian** environments

Absolutism

There is a system of norms and values that should not be questioned

- **No exception:** a rule is a rule
- Typical in **authoritarian** environments

Problems:

- It invalidates moral judging
- Hides the complexity of many dilemmas (e.g. Euthanasia and a murder are not the same type of killing)
- Leaves no room for independent moral judgement

Example: The effort to punish women for having abortions:

<https://www.nytimes.com/2022/08/23/podcasts/the-daily/abortion-abolition-roe-v-wade.html>

- **Both extremes are unsatisfactory**
- Signs of intellectual **laziness**
- Good Moral judgment is often about finding good balance (unbiased)
- This is why it is hard

Argumentation by Analogy

A very useful and common argumentation for moral judgements

Arg. by Analogy

If something is clearly ethical/unethical in an example case, then it is also so for a comparable situation

Critical questions:

- The moral judgement of the example case is clear?
- How comparable are the two situations?
 - similarities are relevant?
 - differences are irrelevant?

Argumentation by Analogy: Example

Example (Hacking)

- Gain access to computer networks that belongs to others
- Hacking somebody's computer is morally wrong?:
 - Arg. not by analogy: motivation, damage, ...
 - Arg. by analogy: breaking into private property

Ethical Dilemmas/ Thought Experiments

Ethical Dilemma

Situation in which a difficult **choice** has to be made between two courses of action, either of which entails transgressing a moral principle.

Thought Experiment

A hypothetical situation in which a hypothesis, theory, or principle is laid out for the purpose of thinking through its consequences

- Ethicists often use hypothetical ethical dilemmas as thought experiments

Trolley Problem, P. Foot 1965

A trolley runs down a track; ahead are five people awaiting certain death. You observe the scenario from nearby and see a lever next to you. If you pull the lever you can divert the trolley to a different set of tracks. Yet, on that other track is a single person. The train cannot be stopped.

Solution to the Trolley Problem

The Trolley Problem is not designed to have a solution. It is, rather, intended to provoke thought, and create an intellectual discourse in which the difficulty of resolving moral dilemmas is appreciated, and our limitations as moral agents are recognized.

- It has recently gained popularity because of self-driving cars.
- It has been used to highlight how culture shapes our morality (<https://www.vox.com/future-perfect/2020/1/24/21078196/morality-ethics-culture-universal-subjective>)

Ethical Dilemmas/ Thought Experiments

Sometimes they help us clarify our thoughts by reasoning by analogy.

Hospital in flames

You are running away from a hospital in flames. You can take with you (i.e, save) babies or in-vitro fetus. What would you do?

The previous Dilemma can be useful to understand intuitions about abortion

Ethical Dilemmas/ thought Experiments/ Reas. by Analogy

The following analysis of **natural abortion** may help you understand your intuitions about **unnatural abortion**

Natural abortion and pain

Who and how much sorry you would feel for:

Ethical Dilemmas/ thought Experiments/ Reas. by Analogy

The following analysis of **natural abortion** may help you understand your intuitions about **unnatural abortion**

Natural abortion and pain

Who and how much sorry you would feel for:

- a woman has a miscarriage a few weeks after pregnancy

Ethical Dilemmas/ thought Experiments/ Reas. by Analogy

The following analysis of **natural abortion** may help you understand your intuitions about **unnatural abortion**

Natural abortion and pain

Who and how much sorry you would feel for:

- a woman has a miscarriage a few weeks after pregnancy
- a woman has a miscarriage 7 months after pregnancy

Ethical Dilemmas/ thought Experiments/ Reas. by Analogy

The following analysis of **natural abortion** may help you understand your intuitions about **unnatural abortion**

Natural abortion and pain

Who and how much sorry you would feel for:

- a woman has a miscarriage a few weeks after pregnancy
- a woman has a miscarriage 7 months after pregnancy
- a baby dies a few days after being born

Ethical Dilemmas/ thought Experiments/ Reas. by Analogy

The following analysis of **natural abortion** may help you understand your intuitions about **unnatural abortion**

Natural abortion and pain

Who and how much sorry you would feel for:

- a woman has a miscarriage a few weeks after pregnancy
- a woman has a miscarriage 7 months after pregnancy
- a baby dies a few days after being born
- a four years old child dies

Ethical Dilemmas/ thought Experiments/ Reas. by Analogy

The following analysis of **natural abortion** may help you understand your intuitions about **unnatural abortion**

Natural abortion and pain

Who and how much sorry you would feel for:

- a woman has a miscarriage a few weeks after pregnancy
- a woman has a miscarriage 7 months after pregnancy
- a baby dies a few days after being born
- a four years old child dies

But of course you have to take thought experiments with a *grain of salt*, because our perceptions/intuition cannot be always taken as the right criteria.

- E.g.: think of a racist version of the trolley problem

Abstract frameworks:

- **Rational** (non-emotional, objective) approach to Ethics
- Try to **characterize** which human actions are morally right/wrong (it should not be arbitrary)
- Try to **guide** humans in making morally right actions (or chose among alternatives)
- Unfortunately, (typically) they do not give (precise) answers
- But help us formulate good questions
- And make our thinking more precise, objective,...

Structure of human action

An **action** is carried out by a certain **actor** which has **consequences**

Structure of human action

An **action** is carried out by a certain **actor** which has **consequences**

Structure of human action

An **action** is carried out by a certain **actor** which has **consequences**

- **Values:** Lasting convictions about the society that we all want (related to consequences)
 - e.g. safe, equal, healthy society

Structure of human action

An **action** is carried out by a certain **actor** which has **consequences**

- **Values:** Lasting convictions about the society that we all want (related to consequences)
 - e.g. safe, equal, healthy society
- **Norms:** Rules that tell what actions are required, permitted or forbidden (related to actions)
 - e.g. obey traffic lights, help a person in need, be reliable at work,...

Structure of human action

An **action** is carried out by a certain **actor** which has **consequences**

- **Values:** Lasting convictions about the society that we all want (related to consequences)
 - e.g. safe, equal, healthy society
- **Norms:** Rules that tell what actions are required, permitted or forbidden (related to actions)
 - e.g. obey traffic lights, help a person in need, be reliable at work,...
- **Virtues:** Desirable human quality (related to actors)
 - e.g. generous, just, empathetic person

E.g., In the **trolley problem**: (**value**: safety), (**norm**: treat all lives as equal or minimize number of casualties), (**virtue**: fairness or compassion)

They are more or less connected

Example: An activist fights for public education

- **Values:** Equal opportunities
- **Norms:** Help disadvantaged people
- **Virtues:** Justice, fairness, solidarity

A fair person (virtue) will want a society with equality of opportunities (value) and therefore help disadvantaged people (norm)

We can classify Ethical Theories as:

- **Consequentialism** (look at consequences-values)
- **Deontology** (look at actions-norms)
- **Virtue Ethics** (look at actors-virtuousity)

We are going to see next the three most influential Theories in Western culture

As we will see, **they are not perfect**, but help us guide our moral judgements

Utilitarianism

Founded by **Jeremy Bentham** (London 1748)

Utilitarianism

- Actions are **judged** by the amount of **pleasure and pain** they bring about.

Utilitarianism

Founded by **Jeremy Bentham** (London 1748)

Utilitarianism

- Actions are **judged** by the amount of **pleasure and pain** they bring about.
- People deserve to be treated ethically because they are **sentient** (have feelings)

Utilitarianism

Founded by **Jeremy Bentham** (London 1748)

Utilitarianism

- Actions are **judged** by the amount of **pleasure and pain** they bring about.
- People deserve to be treated ethically because they are **sentient** (have feelings)

Utility Principle:

Chose the action that brings the greatest happiness for the greatest number of people.

Utilitarianism

Founded by **Jeremy Bentham** (London 1748)

Utilitarianism

- Actions are **judged** by the amount of **pleasure and pain** they bring about.
- People deserve to be treated ethically because they are **sentient** (have feelings)

Utility Principle:

Chose the action that brings the greatest happiness for the greatest number of people.

- It is a **consequentialist** theory where **happiness/well-being** is the value of interest.

Drawback of Utilitarianism

- **Assumption:** pleasure and pain can be measured (moral balance sheet, *felicific calculus* to combine *hedons* and *dolors*)
 - e.g. buy a car or donate to a child hospital
 - e.g. donate to **Wikipedia** or to **Doctors Without Borders**

Drawback of Utilitarianism

- **Assumption:** pleasure and pain can be measured (moral balance sheet, *felicific calculus* to combine *hedons* and *dolors*)
 - e.g. buy a car or donate to a child hospital
 - e.g. donate to **Wikipedia** or to **Doctors Without Borders**
- The assumption is not realistic
- Very problematic in practice
- An extreme trust in this theory (absolutism): the end justifies the means

Drawback of Utilitarianism

- **Assumption:** pleasure and pain can be measured (moral balance sheet, *felicific calculus* to combine *hedons* and *dolors*)
 - e.g. buy a car or donate to a child hospital
 - e.g. donate to **Wikipedia** or to **Doctors Without Borders**
- The assumption is not realistic
- Very problematic in practice
- An extreme trust in this theory (absolutism): the end justifies the means

Utilitarian argumentation usually rely on **cost-benefit** analysis. But simplistic analysis are usually not satisfactory

Utilitarianism: Improvements and Refinements

When different amounts of happiness/pain go to different people, one should take this into account.

Utilitarianism: Improvements and Refinements

When different amounts of happiness/pain go to different people, one should take this into account.

Hierarchy of pleasures, J. Stuart Mill (U.K. 1806)

Some *pleasures* are preferred over others

- E.g., A servant stays outdoors on a rainy day while her master attends the opera. Is not OK

Utilitarianism: Improvements and Refinements

When different amounts of happiness/pain go to different people, one should take this into account.

Hierarchy of pleasures, J. Stuart Mill (U.K. 1806)

Some *pleasures* are preferred over others

- E.g., A servant stays outdoors on a rainy day while her master attends the opera. Is not OK

Harm principle, J. Stuart Mill (U.K. 1806)

Everyone is free to search for his own pleasure as long as he does not harm others. This was a **moral revolution**.

- E.g., A gay couple kissing on the street may offend some people. Is OK

Find here a modern description of Felicities Calculus

Utilitarianism: Improvements and Refinements

In many situations the computation of happiness has to be **adjusted**

- **Uncertainty:** Expected consequences (multiply pleasure by probability, 1 Euro = 2 Euros with 50% chance)
 - donate to a charity feeding poor people or to a research program for malaria vaccine?

Utilitarianism: Improvements and Refinements

In many situations the computation of happiness has to be **adjusted**

- **Uncertainty:** Expected consequences (multiply pleasure by probability, 1 Euro = 2 Euros with 50% chance)
 - donate to a charity feeding poor people or to a research program for malaria vaccine?
- **Distributed Happiness:** Gives value to a just distribution of happiness (2 Euros for 2 people have more value than 4 for 1 person)
 - donate computers to all schools or computers+digital boards + lego robots to some schools?

Utilitarianism: Improvements and Refinements

In many situations the computation of happiness has to be **adjusted**

- **Uncertainty:** Expected consequences (multiply pleasure by probability, 1 Euro = 2 Euros with 50% chance)
 - donate to a charity feeding poor people or to a research program for malaria vaccine?
- **Distributed Happiness:** Gives value to a just distribution of happiness (2 Euros for 2 people have more value than 4 for 1 person)
 - donate computers to all schools or computers+digital boards + lego robots to some schools?
- **Marginal utility:** Same consequences may have different values (e.g. Giving \$1000 to you or to a millionaire)

Kantian Theory

Founded by **Immanuel Kant** (Germany, 1724)

Kantian Theory

- We should be able to determine **norms** telling what is right and wrong through **reasoning**.

Kantian Theory

Founded by **Immanuel Kant** (Germany, 1724)

Kantian Theory

- We should be able to determine **norms** telling what is right and wrong through **reasoning**.
- Actions are judged from how they fit those norms

Kantian Theory

Founded by **Immanuel Kant** (Germany, 1724)

Kantian Theory

- We should be able to determine **norms** telling what is right and wrong through **reasoning**.
- Actions are judged from how they fit those norms
- We should follow the norms out of **duty**

Kantian Theory

Founded by **Immanuel Kant** (Germany, 1724)

Kantian Theory

- We should be able to determine **norms** telling what is right and wrong through **reasoning**.
- Actions are judged from how they fit those norms
- We should follow the norms out of **duty**
- People deserves to be treated ethically because they are **rational beings**

Kantian Theory

Founded by **Immanuel Kant** (Germany, 1724)

Kantian Theory

- We should be able to determine **norms** telling what is right and wrong through **reasoning**.
 - Actions are judged from how they fit those norms
 - We should follow the norms out of **duty**
 - People deserves to be treated ethically because they are **rational beings**
-
- It is a **Deontological** theory: Ethics is about moral rules (norms)

Kantian Theory

Founded by **Immanuel Kant** (Germany, 1724)

Kantian Theory

- We should be able to determine **norms** telling what is right and wrong through **reasoning**.
 - Actions are judged from how they fit those norms
 - We should follow the norms out of **duty**
 - People deserves to be treated ethically because they are **rational beings**
-
- It is a **Deontological** theory: Ethics is about moral rules (norms)
 - **Assumption:** it is possible to make a **universal deontological code** (do you know any deontological code?)

Kantian Theory

Founded by **Immanuel Kant** (Germany, 1724)

Kantian Theory

- We should be able to determine **norms** telling what is right and wrong through **reasoning**.
 - Actions are judged from how they fit those norms
 - We should follow the norms out of **duty**
 - People deserves to be treated ethically because they are **rational beings**
-
- It is a **Deontological** theory: Ethics is about moral rules (norms)
 - **Assumption:** it is possible to make a **universal deontological code** (do you know any deontological code?)
 - **Assumption:** Trust in human kind (reasoning and good will)

Kantian Theory

Kant claims that **ethical guidance** can be identified with the following two principles:

Kant claims that **ethical guidance** can be identified with the following two principles:

- **Universality principle:** act as you would like everybody else to act in similar situations
 - e.g. recycling
 - e.g. cheating in exams

Kant claims that **ethical guidance** can be identified with the following two principles:

- **Universality principle:** act as you would like everybody else to act in similar situations
 - e.g. recycling
 - e.g. cheating in exams
- **Reciprocity Principle:** Treat others as an *end*, not as a *mean* (treat others as you would like others to treat you)
 - We should not *use* (i.e, take advantage of) people
 - Recognizes *dignity* of human beings
 - e.g. misleading advertisement
 - e.g. programmed obsolescence

Kant claims that **ethical guidance** can be identified with the following two principles:

- **Universality principle:** act as you would like everybody else to act in similar situations
 - e.g. recycling
 - e.g. cheating in exams
- **Reciprocity Principle:** Treat others as an *end*, not as a *mean* (treat others as you would like others to treat you)
 - We should not *use* (i.e, take advantage of) people
 - Recognizes *dignity* of human beings
 - e.g. misleading advertisement
 - e.g. programmed obsolescence

Motivation is duty: we should not look after our sick mother out of love, compassion,... but out of duty.

Kantian Theory: Improvements and Refinements

- Can we make a list of **unambiguous** and **consistent** rules?
- Often there are contradictory norms

Kantian Theory: Improvements and Refinements

- Can we make a list of **unambiguous** and **consistent** rules?
- Often there are contradictory norms
- To cope with this problem: norms are arranged as a **hierarchy**. Rules are to be followed unless it is inconsistent with a more important rule.
Example: I promise you that I will correct the exam within a week. Then, a friend asks me for help and I need to delay my correction.

Kantian Theory: Improvements and Refinements

- Can we make a list of **unambiguous** and **consistent** rules?
- Often there are contradictory norms
- To cope with this problem: norms are arranged as a **hierarchy**. Rules are to be followed unless it is inconsistent with a more important rule.
Example: I promise you that I will correct the exam within a week. Then, a friend asks me for help and I need to delay my correction.
- Another way to cope with this problem is to make **domain-specific** lists instead of general-purpose.
Example: Professional deontological codes, Human Rights Declaration,...

Virtue Ethics

Founded by **Aristotle** (Greece, 384 BC)

Virtue Ethics

- There are some **personality traits** (called **virtues**) that define **morally good people**.

Founded by **Aristotle** (Greece, 384 BC)

Virtue Ethics

- There are some **personality traits** (called **virtues**) that define **morally good people**.
- When confronted with a moral dilemma ask "what should I be"? (instead of "what should I do?")

Founded by **Aristotle** (Greece, 384 BC)

Virtue Ethics

- There are some **personality traits** (called **virtues**) that define **morally good people**.
- When confronted with a moral dilemma ask "what should I be"? (instead of "what should I do?")
- Virtuous people live a **good life**. Their life is meaningful and it makes them happy (a very special type of happiness)

Founded by **Aristotle** (Greece, 384 BC)

Virtue Ethics

- There are some **personality traits** (called **virtues**) that define **morally good people**.
- When confronted with a moral dilemma ask "what should I be"? (instead of "what should I do?")
- Virtuous people live a **good life**. Their life is meaningful and it makes them happy (a very special type of happiness)
- An action is **morally acceptable** if that action is what a virtuous person would do.

Recommendation: *The Good Life* (available in Netflix)

- **Assumption:** We can identify or imagine virtuous people and be inspired by them

- **Assumption:** We can identify or imagine virtuous people and be inspired by them
- Virtues can be classified:
 - **Intellectual Virtues:** wisdom, rationality, ...
 - **Moral Virtues:** fairness, benevolence, honesty, loyalty, courage, prudence,...
 - **Professional Virtues:** (will depend on the profession)

- **Assumption:** We can identify or imagine virtuous people and be inspired by them
- Virtues can be classified:
 - **Intellectual Virtues:** wisdom, rationality, ...
 - **Moral Virtues:** fairness, benevolence, honesty, loyalty, courage, prudence,...
 - **Professional Virtues:** (will depend on the profession)
- **Golden Mean:** Virtues require a balance between two behavioural extremes (e.g. courage is the balance between foolhardiness and cowardice)

- **Assumption:** We can identify or imagine virtuous people and be inspired by them
- Virtues can be classified:
 - **Intellectual Virtues:** wisdom, rationality, ...
 - **Moral Virtues:** fairness, benevolence, honesty, loyalty, courage, prudence,...
 - **Professional Virtues:** (will depend on the profession)
- **Golden Mean:** Virtues require a balance between two behavioural extremes (e.g. courage is the balance between foolhardiness and cowardice)
- The relevance of some virtues may be context-dependent.
 - Virtuous **parent:** care-taker, loving, generous.
 - Virtuous **employee:** responsible, loyal, trustable.
 - Virtuous **engineer:** techn. enthusiast, socially responsible, honesty, confidentiality,

Example

Is lying morally wrong?

Example

Is lying morally wrong?

From a Virtue Ethics perspective, you should not ask whether lying is wrong. You should ask whether lying is virtuous or not.

Example

Is lying morally wrong?

From a Virtue Ethics perspective, you should not ask whether lying is wrong. You should ask whether lying is virtuous or not.

In the case of lying, it is context dependent

- E.g. Lying not to hurt feelings is clearly virtuous

Summary

	Utilitarianism	Kantian	Virtue
Look at	Conseq. confronted w. values (happin)	Actions confronted with norms	Actors confronted with virtues
Why	"victims" are sentient	"victims" are rational	we want meaningful life
Tool	Moral Balance Sheet	Universality and Recipr. Principles	Virtuous People
Motiv.	Max happiness	Duty	Good life

Disclaimer

We are going to apply this theories to some examples.

Recall that the theories do not give answers, just an approach to deal to the problem.

In most cases the same theory can be used to reach contradictory conclusions

Example: organ sales (as opposed to organ donation)

Write one argumentation either in favor or against for each one of the three Ethical Theories that we have studied.

By organ sales or donation we consider organs that are not needed by the seller or the donor (e.g. kidney or after death)

Example: organ sales (as opposed to organ donation)

Write one argumentation either in favor or against for each one of the three Ethical Theories that we have studied.

By organ sales or donation we consider organs that are not needed by the seller or the donor (e.g. kidney or after death)

FYI:

- Currently, organs can only be donated
- There is a clear shortage of donors

Utilitarian

Clearly, organ donation makes organs very scarce and many people die because they cannot have a transplant.

If we allow organ sales, many more people would make their organs available so more lives would be saved. Besides, a free market would set a fair price, so the sellers would only accept a reasonable price for them, which means that they would also be satisfied with the deal. Therefore, organ sales is a win-win system that has better consequences than organ donation

Kantian

Because the world has important economical inequalities, organ sales would make poor people sell their organs to rich people. In many cases, out of desperation. Poor people would not have access to organs even if they really needed them. In that context, organ sellers would be treated as a means, which is unacceptable.

On the contrary, if you think of the Universality and Reciprocity principles, people has the duty to donate their organs (it is morally wrong not to do it).

Virtue Ethics

Here are many virtues that we can associate to organ donation: altruism, generosity, empathy,...

A virtuous person would think that donating organs makes her life more meaningful and joyful. That is, a better life.

On the contrary, selling your organs is not virtuous at all.

Example: deadline extension

A student does not hand in a deliverable in time. After the deadline, he asks very politely if he can have an extension. Should I grant the extension?

Example: deadline extension

A student does not hand in a deliverable in time. After the deadline, he asks very politely if he can have an extension. Should I grant the extension?

- **Utilitarianism:**

- He will be happier
- He will not drop the course (good for his future and aligned with my teaching goals)
- Nobody is hurt by the extension

Example: deadline extension

A student does not hand in a deliverable in time. After the deadline, he asks very politely if he can have an extension. Should I grant the extension?

- **Utilitarianism:**

- He will be happier
- He will not drop the course (good for his future and aligned with my teaching goals)
- Nobody is hurt by the extension

- **Kant:**

- the deadline was known well in advance, and deadlines are to be followed
- all students should be treated equally.

Example: deadline extension

A student does not hand in a deliverable in time. After the deadline, he asks very politely if he can have an extension. Should I grant the extension?

- **Utilitarianism:**

- He will be happier
- He will not drop the course (good for his future and aligned with my teaching goals)
- Nobody is hurt by the extension

- **Kant:**

- the deadline was known well in advance, and deadlines are to be followed
- all students should be treated equally.

- **Virtue:** conflict between benevolence and fairness

- because this conflict happens in the professional sphere, I think fairness should prevail.

Example: Bank petty theft

Suppose you work for a bank with 10 million customers. You know that nobody notices if their balance decreases by 1 cent. You are tempted to steal 1 cent to each customer to cover your son's college expenses

Example: Bank petty theft

Suppose you work for a bank with 10 million customers. You know that nobody notices if their balance decreases by 1 cent. You are tempted to steal 1 cent to each customer to cover your son's college expenses

- **Utilitarianism:** if nobody notices, there is no harm
- **Kant:** it attempts against both universality and reciprocity principles
- **Virtue:** A virtuous employee should be loyal, trustable,... so she would not do that

Example: torture

Suppose you arrested a terrorist that has just put a bomb in an unknown place. Should you torture him to find out where the bomb is?

- **Utilitarianism:** Torture may be right when you can achieve a greater good
- **Kant:** Torture is always wrong? (Universality principle). Does torture respect dignity? (Reciprocity principle)
- **Virtue:** Different virtues are in conflict here. Prudence (is torture the only way?), Rationality (will torture work?), benevolence, compassion, fairness, ...

Example

TV broadcast

Imagine that you are a technician providing the signal during the soccer world cup (300 million spectators) and you witness a heart attach. Your choice: keep the broadcasting or save the person

- Utilitarianism:
- Kant:
- Virtue:

Making Ethical Decisions

Example:

Imagine that you are a technician providing the signal during the soccer world cup (300 million spectators) and you witness a heart attack. Your choice: keep the broadcasting or save the person

- **Utilitarianism:**

- Watching game: $10 \times 3 \times 10^8$, saving life 2×10^5 (from insurance company)
- live is (infinitely?) more important than sport
- ...

- **Kant:** "saving a life" is a rule that you want to be universal? Do we want a world where massive entertainment is more important than life? (hunger games)

- **Virtue:** kindness (general human virtue) vs loyalty (professional virtue)

One of the most successful cars in the 70s was the Ford Pinto. The main reason for its success was its very attractive price of \$2000 (end-range cars at the time typically doubled this price). As could not be otherwise, its design passed all safety requirements at the time. The investigation of a very unfortunate accident in which 3 teenagers died brought to light that the accident was deadly because the car was hit by a truck from behind and the car exploded. Such explosion could have been avoided by a change in the car design that would have added an extra cost of \$11 to the manufacturing of each unit.

At the time of its design, the expected number of sales for Ford Pinto was estimated as 11 million units. An educated guess would predict that around 100 out of all the Ford Pinto on the road would be involved in a rear-end accident similar to the fatal one. The vision statement of the Ford company is "*People working together as a lean, global enterprise to make people's lives better through automotive and mobility leadership.*"

Put yourself on the shoes of the Ford company and justify **in three different ways** why there was nothing morally wrong in the design of the car. Each one of the justifications must be based on one of the three ethical theories that we have studied in class.

Utilitarianism:

Affordable cars clearly improve the life of people (that is, bring happiness). Driving a car, by definition, has a risk and it is obvious that a cheap car will be less safe. An utilitarian argument would compare the "happiness" given by having a Ford Pinto (may be compared with having an expensive safer car) with the "added risk of accident" of a Ford Pinto compared with that of an expensive car. Of course, we have very little information and our argument will be simplistic, but we can put both elements in a common currency for comparison purposes as follows: The "benefit" of driving a Ford Pinto w.r.t. an expensive car can be valued as \$2000 (the money that you save). The (uncertain) "cost" of driving a Ford Pinto w.r.t. an expensive car is the product of the probability of having a fatal accident (for instance, $100/11000000$) the average number of passengers (for instance, 2) and the cost of a human life (for instance taking it from insurance companies and let it be \$100000) which is roughly \$2. With this data, the benefits clearly outweigh the cost.

Kantian:

From the vision statement of Ford we can somehow imagine the ethical norms (deontology) enforced at the company with respect to their products. Their goal is to improve people's life by improving their mobility (through cars). Designing the affordable (and still quite safe) Ford Pinto is clearly aligned with the company's goal and cannot go against its ethical norms. As long as the car passed all safety tests required by the law, and the engineers designed it with care and professionalism, no norm is broken. Besides, we shouldn't underestimate the intelligence of potential buyers. Ford offers a variety of models, and buyers are free to decide if they want to pay or not the additional cost of a safer car (larger, heavier, tougher, with wider tires,...)

Virtue Ethics:

There are many virtues that we would like companies to have (or company CEOs to enforce). For the purpose of this exercise we can identify that there is a tension between compassion (the virtue of caring about others' safety) and technology enthusiasm (the virtue of willing to improve peoples lives through technological progress). Since these two virtues are in conflict, a virtuous company should attain a good balance (golden mean, in Aristotle's words). Since we know that the price of the car was very attractive and we only have information about one particular questionable design decision and it is very infrequent, we can claim that the design of the car was done balancing well the two virtues.

Are Ethical Dilemmas Thought Experiments?

Self-driving Car Problem

You are a Software Engineering programming a self-driving car

Are Ethical Dilemmas Thought Experiments?

Self-driving Car Problem

You are a Software Engineering programming a self-driving car

I think this is a wrong way to approach the problem. As engineers, we should focus on safe systems...

We will get back to this when we talk about deontological codes for engineers.

Example

- **Kid:** I do not want to eat my vegetables
- **Mother:** Think of all the poor kids in the world that have no food
- **Kid:** How do they care about what I do?



Example

- The mother turns the discussion into an moral discussion (she could have threaten the kid with a punishment, instead)

Example

- The mother turns the discussion into an moral discussion (she could have threaten the kid with a punishment, instead)
- The kid is being Utilitarian (the consequence of throwing away food do not cause any harm)
- What about the mother?:
 - Throwing away your food is wrong? (virtue ethics)
 - Throwing away food may be against the dignity of people in need (kantian ethics)

Example

- The mother turns the discussion into an moral discussion (she could have threaten the kid with a punishment, instead)
- The kid is being Utilitarian (the consequence of throwing away food do not cause any harm)
- What about the mother?:
 - Throwing away your food is wrong? (virtue ethics)
 - Throwing away food may be against the dignity of people in need (kantian ethics)
 - She is really lying to you: she cares about being good care-takers (virtue)
 - By lying, is she treating you with dignity? (Kant)

Peter Singer has made this fun toy to play with: **Peter Singer's Bot** You can ask questions such as:

- how much money should I give to a charity?
- should I tell my girlfriend that I cheated on her?
- is it ok to cheat on exams?